State prep [bonus]

Saturday, July 3, 2021 2:00 PM

Introduction to quantum noise

State preparation noise

Qiskit Global Summer School on Quantum Machine Learning

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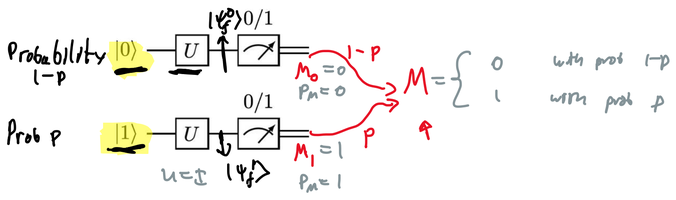


Density operator

$$\rho = (1-p)\,|0\rangle\langle 0| + p\,|1\rangle\langle 1|$$
$$= p_{\text{no error}}\,\rho_{\text{ideal}} + p_{\text{error}}\,\rho_{\text{error}}$$

$$\rho_f = p_{\text{no error}}\,\rho_{\text{ideal}}^{\text{final}} + p_{\text{error}}\,\rho_{\text{error}}^{\text{final}}$$
$$= (1-p)\,|\psi_f^0\rangle\langle\psi_f^0| + p\,|\psi_f^1\rangle\langle\psi_f^1|$$
$$= (1-p)\,U|0\rangle\langle 0|U^\dagger + p\,U|1\rangle\langle 1|U^\dagger$$
$$= U\left[(1-p)\,|0\rangle\langle 0| + p|1\rangle\langle 1|\right]U^\dagger$$
$$= U\,\rho\,U^\dagger$$

$(U|\psi\rangle)^\dagger = (|\psi_f\rangle)^\dagger$

$\langle\psi_f| = \langle\psi|U^\dagger$

$$\rho_U = U\rho U^\dagger$$
$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$
$$|1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$
$$\rho = \begin{pmatrix} 1-p & 0 \\ 0 & p \end{pmatrix} = \frac{1}{2}\left(\hat{I} + (1-2p)\hat{Z}\right)$$

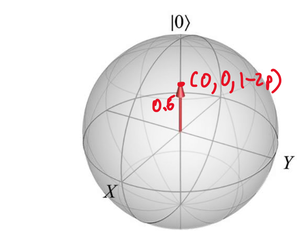
$\langle 0|$ $\langle 1|$

$\langle X\rangle = \mathrm{Tr}[X\rho] = 0$

$\langle Y\rangle = \mathrm{Tr}[Y\rho] = 0$

$\langle Z\rangle = \mathrm{Tr}[Z\rho] = 1-2p$

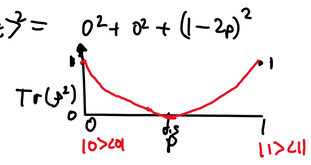
$\mathrm{Tr}[P_\alpha P_\beta] = 2\delta_{\alpha\beta}$ for $\alpha, \beta \in \{X, Y, Z\}$



$p = 0.2$ 20% to be $|1\rangle$

$= \left(\uparrow 1\right) \times 1-p + \left(\downarrow 1\right) \times p$

$\mathrm{Tr}(\rho^2) = \langle X\rangle^2 + \langle Y\rangle^2 + \langle Z\rangle^2 = 0^2 + 0^2 + (1-2p)^2$



Scaling to larger number of qubits

$[(1-p)\,|0\rangle\langle 0| + p\,|1\rangle\langle 1|]^{\otimes n}$ — U

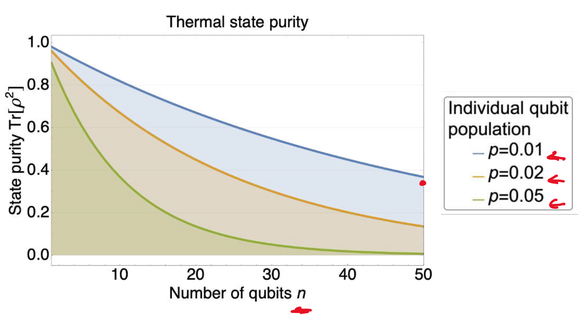
$\rho_0 = \left[(1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|\right]^{\otimes n}$

$= (1-p)^n |0000\cdots 0\rangle\langle 0\cdots 0| + \cdots + |10100\cdots\rangle\langle\cdots|$

$$P(M = 0000\cdots 0) = \begin{cases} 1 & \text{if } p = 0 \quad \text{(ideal case)} \\ (1-p)^n \simeq 1 - np + O(p^2) & \text{for } p > 0 \end{cases}$$

$\langle Z_k\rangle = \langle \underset{\text{qubit } k}{|||\,Z\,|||}\cdots\rangle = 1-2p$ — much faster

$\langle ZZZ\cdots Z\rangle = \mathrm{Tr}\left[Z^{\otimes n}\rho\right] = (1-2p)^n$

$$\rho_0 = \prod_{k=1}^{n} \otimes \frac{1}{2}\left(\hat{I}_k + (1-2p)\hat{Z}_k\right)$$
$$= \frac{1}{2^n}(\mathbb{1} + Z) \otimes (\mathbb{1} + Z) \otimes \cdots$$
$$= \frac{1}{2^n}\left[(1-2p)^n Z^{\otimes n} + \cdots\right]$$

$$\mathrm{Tr}[\rho_0^2] = \prod_{k=1}^{n} \mathrm{Tr}(\rho_{0k}^2) = (1-2p)^{2n}$$